THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A farmer grows wheat after pulses and applies less urea. The reason is: 1

(a) wheat needs no nitrogen **(b)** pulses consume nitrogen
(c) nitrogen left by pulses **(d)** urea is costly

2. A mirror reflects the colours of objects. A red rose in white light viewed in a plane mirror appears: 1

(a) red **(b)** green **(c)** white **(d)** black

3. A student claims 'the Sun is closer to Earth in summer'. A good test is: 1

(a) measure the temperature once
(b) look at the Sun
(c) ask friends
(d) compare dates of nearest distance with seasons

4. In a set-up with a plant in a bell jar with NaOH solution, starch is absent in the leaf because: 1

(a) NaOH adds oxygen **(b)** NaOH absorbs carbon dioxide
(c) light is blocked **(d)** NaOH gives CO2

5. A person 1.6 m tall wishes to see a full-length view in a plane mirror. The minimum mirror length is: 1

(a) 3.2 m **(b)** 1.6 m **(c)** 0.8 m **(d)** 0.4 m

6. If the Earth revolved around the Sun in half the time, a year would have about: 1

(a) 730 days **(b)** 182 days **(c)** 91 days **(d)** 365 days

7. Which of these is a net result of photosynthesis in a day? 1

(a) only oxygen **(b)** less glucose
(c) more glucose than used in respiration **(d)** no glucose

8. A student sees a shadow of her hand on a screen 2 m from a lamp. If she moves the hand to the midpoint between lamp and screen, the shadow size relative to the hand is: 1

(a) four times **(b)** same **(c)** half **(d)** double

9. The Sun is much larger than the Moon, yet in a total solar eclipse the Moon can hide it because the Moon is: 1

(a) larger **(b)** much nearer to the Earth
(c) brighter **(d)** much hotter

10. A student claims that mushrooms make food in sunlight. The correct reply is: 1

(a) mushrooms have no chlorophyll and absorb food from dead matter
(b) correct
(c) they use CO2
(d) they make starch

11. The colour of the shadow of a red ball in white light is: 1

(a) white **(b)** green **(c)** black **(d)** red

12. A student says that 'the Moon has phases because the Earth's shadow falls on it'. This is: 1

(a) correct for the first quarter **(b)** correct
(c) incorrect **(d)** partly correct

13. In an experiment, aquatic plant produces 10 bubbles per minute at 30 cm from a lamp and 40 at 10 cm. This shows photosynthesis rate: 1

(a) increases with light intensity **(b)** is independent of light
(c) stops at night **(d)** decreases with light

14. A girl walks away from a mirror at 2 m/s for 3 s. How far has the distance between her and her image increased? 1

(a) 2 m **(b)** 12 m **(c)** 6 m **(d)** 3 m

15. If the Earth were upright (no tilt), the Sun at noon on the equator would be: 1

(a) overhead only in June **(b)** never overhead
(c) overhead twice **(d)** overhead all year

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why do the Sun and the Moon rise in the east and set in the west? 2

17. How do plants adapt to deserts to reduce transpiration? 2

18. Explain why the Moon looks bright at night but is not a luminous body. 2

19. Why do places at high latitudes have very long summer days? 2

20. Explain why the leaf in the dark does not show starch though it is green. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Why do you see your face in a steel spoon but not as clearly as in a mirror? 3

22. Discuss how the study of eclipses helped scientists, with two examples. 3

23. Explain the changes in a leaf's starch content from morning to night and the reason. 3

24. A girl is 4 m from a mirror and walks 1 m towards it. By how much does the distance between her and her image decrease? 3

OR

If the new moon occurs on 1 March, estimate the dates of first quarter, full moon and last quarter. 3

25. Explain with examples how plants help in controlling global warming. 3

OR

Why are reflecting surfaces on roads and bicycles used? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Compare a pinhole camera and a plane mirror as image forming devices. 5

OR

Explain how astronomers predict the time of eclipses using the motions of the Sun, Earth and Moon. 5

27. A plant biologist finds a plant with no chlorophyll which grows on the bark of a tree. Discuss the possible modes of nutrition and how to decide. 5

OR

Design an experiment to find the relation between the object distance and the image distance in a plane mirror, with a table and conclusion. 5

END OF QUESTION PAPER Total: 50 Marks

MASTERMIND